ELECTRICAL MACHINES · PAPER

Jun 2024 exam: full worked solutions

Answers and worked solutions · Day 21 · Fri 13 Nov · pack MX2

Question 1. AC and DC Power Circuit Analysis [20 marks]

1a) Fill-In-The Blank Questions [4 marks]

(i) The turns ratio is $n = N_1/N_2 = V_1/V_2$.

- $240/24$: $n = \mathbf{10}$ (10:1), **step down**.
- $240/480$: $n = \mathbf{0.5}$ (1:2), **step up**.

(ii)

$$P_{OUT} = P_{IN} - P_{LOSS} = 1000 - 300 = \mathbf{700\ W}, \qquad \eta = \frac{700}{1000} = \mathbf{70\,\%}$$

- More loss at the same input: η **decreases**.
- More input at the same loss: η **increases**, because the fixed loss becomes a smaller fraction of the input. For example, at 2000 W input, $\eta = 1700/2000 = 85\,\%$.

All answers match the course.

1b) Electrical and Mechanical Power [3 marks]

(i) The 415 V and 6.8 A are line values.

$$S = \sqrt{3}\,V_L I_L = \sqrt{3} \times 415 \times 6.8 = 4888\ \text{VA}$$

$$\text{pf} = \frac{P}{S} = \frac{4000}{4888} = \mathbf{0.8184\ lagging}$$

(ii)

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \times 1420}{60} = 148.7\ \text{rad/s}$$

$$P_{OUT} = \tau\omega = 23 \times 148.7 = \mathbf{3.420\ kW}, \qquad \eta = \frac{3420}{4000} = \mathbf{85.50\,\%}$$

"4 pole, 50 Hz" is not needed here ($n_S = 1500$ rpm). All answers match the course.

1c) AC Circuit Analysis [5 marks]

Figure values. The paper's figure gives $R = 100\ \Omega$ and $jX_L = j40\ \Omega$. As a check, the official answers ($P_R = 400$ W and $Q_L = 160$ VAr at 2 A) give $R = 400/2^2 = 100\ \Omega$ and

$X_L = 160/2^2 = 40\ \Omega$.

(i) In a series circuit the same current flows through both elements.

Element	P	Q
Resistor	$I^2R = 2^2 \times 100 = \mathbf{400\ W}$	$\mathbf{0\ VAr}$
Inductor	$\mathbf{0\ W}$	$I^2X_L = 2^2 \times 40 = \mathbf{160\ VAr}$

(ii) V_R is in phase with I_1, and V_L leads I_1 by 90°. So the two phasors are perpendicular:

$$|V_1| = \sqrt{V_R^2 + V_L^2} = \sqrt{50^2 + 20^2} = \mathbf{53.85\ V}$$

(This is consistent with the figure values: $I_1 = 50/100 = 0.5$ A, so $V_L = 0.5 \times 40 = 20$ V.)

All answers match the course.

1d) Power Factor Correction [8 marks]

(i) [2] Power triangle: horizontal side $P = 80$ W, vertical side $Q = 60$ VAr (upwards, because the load is inductive), hypotenuse $S = 100$ VA. The angle between P and S is $-\phi = 36.87^\circ$.

$$S = \sqrt{80^2 + 60^2} = 100\ \text{VA}, \qquad \text{pf} = \frac{P}{S} = \frac{80}{100} = \mathbf{0.8\ lagging}$$

(ii) [4] An ideal capacitor absorbs no real power, so the supply real power is unchanged:

$$P_S = \mathbf{80\ W}$$

At 0.9 lagging, $-\phi = \cos^{-1} 0.9 = 25.84^\circ$:

$$Q_S = P_S \tan(25.84^\circ) = 80 \times 0.4843 = \mathbf{38.75\ VAr}$$

Conservation of reactive power: $Q_S = Q_{LOAD} + Q_C$, so

$$Q_C = 38.75 - 60 = \mathbf{-21.25\ VAr}$$

The negative sign means the capacitor generates reactive power.

(iii) [2] Use the new value, $Q_C = -30$ VAr. The capacitor sits directly across 115 V.

$$|Q_C| = \frac{V^2}{X_C} = \omega C V^2 \ \Rightarrow\ C = \frac{|Q_C|}{\omega V^2} = \frac{30}{2\pi \times 60 \times 115^2} = \mathbf{6.017\ \mu F}$$

All answers match the course.

Question 2. Magnetic Circuits and Transformers [20 marks]

2a) Electromagnetics [7 marks]

$$A = \pi r^2 = \pi(0.0125)^2 = 4.909 \times 10^{-4}\ \text{m}^2$$

(i) [3] The core has infinite μ, so $\mathcal{R}_{core} = 0$ and only the gap counts.

$$\mathcal{R}_g = \frac{l_g}{\mu_0 A} = \frac{0.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 4.909 \times 10^{-4}} = \mathbf{810.6\ kA/Wb}$$

$$L = \frac{N^2}{\mathcal{R}_g} = \frac{120^2}{810.6 \times 10^3} = \mathbf{17.77\ mH}$$

(ii) [2] Back-emf equation (no reluctance needed):

$$V_{max} = 4.44\, f N \hat{B} A = 4.44 \times 50 \times 120 \times 1.5 \times 4.909 \times 10^{-4} = \mathbf{19.62\ V\ rms}$$

(With the exact factor $2\pi/\sqrt{2} = 4.443$ you get 19.63 V. Both are fine.)

(iii) [2] One gap, so one area:

$$F = \frac{B^2 A}{2\mu_0} = \frac{1.5^2 \times 4.909 \times 10^{-4}}{2 \times 4\pi \times 10^{-7}} = \mathbf{439.5\ N}$$

All answers match the course.

2b) Ideal Transformers [5 marks]

$n = 6$, so $n^2 = 36$.

(i) [2]

$$Z_1 = \mathbf{0.3\ \Omega}, \qquad Z_2 = n^2 Z_1 = 36 \times 0.3 = \mathbf{10.8\ \Omega}, \qquad Z_3 = 15 + 10.8 = \mathbf{25.8\ \Omega}$$

(ii) [3]

$$I_1 = \frac{V_S}{Z_3} = \frac{12}{25.8} = \mathbf{0.4651\ A}, \qquad I_2 = nI_1 = 6 \times 0.4651 = \mathbf{2.791\ A}$$

$$V_1 = I_1 Z_2 = 0.4651 \times 10.8 = \mathbf{5.023\ V}, \qquad V_2 = \frac{V_1}{n} = \mathbf{0.8372\ V}$$

CHECK: $V_1 I_1 = V_2 I_2 = 2.336$ W, as it must be for an ideal transformer. All answers match the course.

2c) Three-Phase Transformers [8 marks]

(i) [2] Simplified circuit, per phase, referred to the primary. Draw the shunt branch $R_C = 250\ \Omega$ in parallel with $jX_M = j14\ \Omega$ at the input terminals. Then draw one series branch:

$$R = R_1 + R_2' = 0.14\ \Omega, \qquad jX = j(X_1 + X_2') = j0.28\ \Omega$$

Label it $0.14\ \Omega$ and $j0.28\ \Omega$, leading to the V_2' terminals.

(ii) [4] Assume star. Per phase:

$$V_1 = \frac{16\,000}{\sqrt{3}} = 9238\ \text{V}$$

On open circuit $I_2' = 0$, so only the shunt branch carries current:

$$Z = 250 \parallel j14 = \frac{1}{\frac{1}{250} + \frac{1}{j14}} = 13.98\angle 86.79^\circ\ \Omega$$

$$|I_1| = \frac{9238}{13.98} = \mathbf{660.9\ A}$$

$$P = 3\frac{V_1^2}{R_C} = 3 \times \frac{9238^2}{250} = \mathbf{1.024\ MW}$$

> **CHECK:** $I_R = 36.95$ A and $I_M = 659.8$ A, and $\sqrt{36.95^2 + 659.8^2} = 660.9$ A. The rated current is 5413 A, so the OC current is 12.2 % of rated.

(iii) [2] The circuit is linear and Z is fixed.

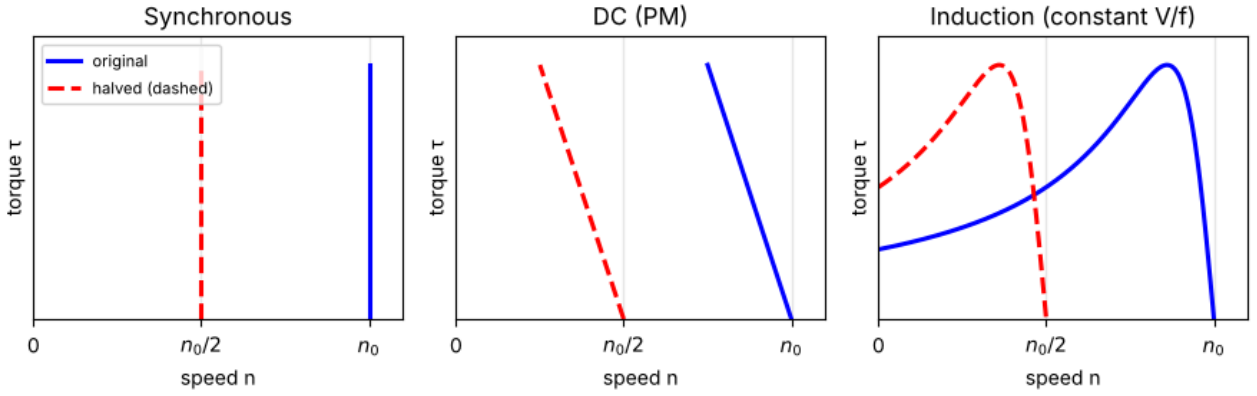
Quantity	Law	Factor
$\lvert I\rvert$	$V/\lvert Z\rvert$	**×0.5**
P	$3V^2/R_C$	**×0.25**
Q	$3V^2/X_M$	**×0.25**
pf	$\cos\angle Z$, independent of V	**×1**

All answers match the course.

Question 3. Electric Machines [20 marks]

3a) Torque vs. Speed Characteristics [6 marks]

The answer sheet names the curves, in order: **synchronous, DC, induction** (Lecture Sections 4.1 and 7.9).



τ–n curves (solid) and the curves after halving the no-load or synchronous speed (dashed). The induction curve is drawn for constant V/f with R_1 neglected.

(i) [3] Name and sketch.

1. **Synchronous machine:** a vertical line at n_S. Halved, it becomes a vertical line at $n_S/2$.
2. **DC machine:** a straight line falling from n_{NL} at $\tau = 0$. Halved, it is a **parallel** line starting at $n_{NL}/2$. The slope (R/k^2) is unchanged.
3. **Induction machine:** zero torque at n_S, a steep, nearly linear region just below n_S, a peak (τ_{max}) and then the starting torque at $n = 0$. Halved at constant V/f, the curve keeps its shape and shifts left to start at $n_S/2$. The slip speed at a given torque and the peak torque are unchanged.

(ii) [3] How to halve the speed safely.

- **Synchronous:** halve f, which halves $n_S = 60f/p$. **Also halve** V, so that V/f (and hence the peak flux) stays constant. Otherwise the core saturates and the current becomes very large.
- **DC: halve the armature voltage** V. The no-load speed is V/k.
- **Induction: halve both** f **and** V (constant V/f), for the same reason as the synchronous machine.

AC machines need a variable-voltage, variable-frequency inverter for this. A DC machine needs a variable DC supply.

3b) Induction Machine Parameters [4 marks]

$R_1 = \mathbf{2\ \Omega}$. It is given per phase, so no halving is needed.

The blocked-rotor test gives the series branch. Per phase:

$$V_1 = \frac{80}{\sqrt{3}} = 46.19\ \text{V}, \qquad I_1 = 5\ \text{A}, \qquad P_1 = \frac{390}{3} = 130\ \text{W}$$

$$R_1 + R_2' = \frac{P_1}{I_1^2} = \frac{130}{25} = 5.2\ \Omega \ \Rightarrow\ R_2' = 5.2 - 2 = \mathbf{3.2\ \Omega}$$

$$S_1 = 46.19 \times 5 = 230.9\ \text{VA}, \quad Q_1 = \sqrt{230.9^2 - 130^2} = 190.9\ \text{VAr}, \quad X = \frac{Q_1}{I_1^2} = \mathbf{7.635\ \Omega}$$

If the split is wanted: $X_1 = X_2' = 3.818\ \Omega$.

The no-load test is not needed for these three. (It gives $R_C = 637.9\ \Omega$ and $X_M = 68.85\ \Omega$.)

All answers match the course.

3c) Induction Machine Calculation [10 marks]

(i) [3] The rated speed of 960 rpm sits just below a synchronous speed. $p = 60f/n = 3000/960 = 3.125$, so $p = 3$ pole-pairs (6 poles).

$$n_S = \frac{60f}{p} = \frac{60 \times 50}{3} = \mathbf{1000\ rpm}, \qquad \omega_S = 104.72\ \text{rad/s}$$

$$s = \frac{n_S - n}{n_S} = \frac{1000 - 1025}{1000} = \mathbf{-0.025}, \qquad \frac{R_2'}{s} = \frac{3}{-0.025} = \mathbf{-120\ \Omega}$$

(ii) [5] Circuit: V_1, then $R_1 = 2.5\ \Omega$, then $jX = j12\ \Omega$, then $R_2'/s = -120\ \Omega$, all in series. There is no shunt branch.

$$V_1 = \frac{415}{\sqrt{3}}\angle 0^\circ = 239.6\angle 0^\circ\ \text{V}$$

$$Z = 2.5 + j12 - 120 = -117.5 + j12 = 118.1\angle 174.2^\circ\ \Omega$$

$$I_1 = \frac{239.6\angle 0^\circ}{118.1\angle 174.2^\circ} = \mathbf{2.029\angle -174.2^\circ\ A}$$

$$P_{IN} = 3V_1 I_1 \cos\phi = 3 \times 239.6 \times 2.029 \times \cos(-174.2^\circ) = \mathbf{-1451\ W}$$

$P_{IN} < 0$, so 1.451 kW flows into the supply: the machine is generating. ($Q_{IN} = +148$ VAr, so it still absorbs reactive power.)

(iii) [2]

$$P_{AG} = 3I_2'^2 \frac{R_2'}{s} = 3 \times 2.029^2 \times (-120) = \mathbf{-1.482\ kW}$$

$$\tau = \frac{P_{AG}}{\omega_S} = \frac{-1482}{104.72} = \mathbf{-14.15\ Nm}$$

The negative torque opposes the prime mover.

Power-flow check: shaft input $= |P_{AG}|(1-s) = 1519$ W. Rotor copper loss $= 3I^2R_2' = 37$ W. Airgap $= 1482$ W. Stator copper loss $= 3I^2R_1 = 31$ W. Electrical output $= 1451$ W, which equals $|P_{IN}|$.

Against the course: all agree. Unrounded, $P_{AG} = -1481.5$ W, which rounds to −1.481 kW. The course's −1.482 kW uses the rounded $I = 2.029$ A ($3 \times 2.029^2 \times 120 = 1482.1$ W). Either gets the mark. The torque is −14.15 Nm either way. (Pack M9 reproduced the same values.)

Question 4. Synchronous Machines [20 marks]

4a) Synchronous Machines [5 marks]

(i) [3] The OC voltage is a line value. At the same field current:

$$|Z_S| = \frac{E_{0,ph}}{I_{SC}} = \frac{12\,000/\sqrt{3}}{6500} = 1.066\ \Omega$$

$$X_S = \sqrt{|Z_S|^2 - R^2} = \sqrt{1.066^2 - 0.2^2} = 1.047\ \Omega$$

$$Z_B = \frac{V_B^2}{S_B} = \frac{(14\times10^3)^2}{400\times10^6} = 0.4900\ \Omega, \qquad X_S = \frac{1.047}{0.49} = \mathbf{2.137\ pu}$$

(ii) [2]

$$\mathbf{V} = \mathbf{0.97\angle 0^\circ\ pu}$$

$$P = \frac{200}{400} = 0.5\ \text{pu}, \quad |S| = \frac{0.5}{0.8} = 0.625\ \text{pu}, \quad |I| = \frac{0.625}{0.97} = 0.6443\ \text{pu}$$

Lagging, so $\phi = -\cos^{-1}0.8 = -36.87^\circ$:

$$\mathbf{I} = \mathbf{0.6443\angle -36.87^\circ\ pu}$$

All answers match the course.

4b) Synchronous Machines [6 marks]

(i) [3] $\mathbf{V} = 1\angle 0^\circ$ pu. Rated VA means $|S| = 1$ pu, so $|I| = 1$ pu. Leading, so $\phi = +\cos^{-1}0.9 = +25.84^\circ$:

$$\mathbf{I} = 1\angle +25.84^\circ\ \text{pu}$$

Generator convention:

$$\mathbf{E}_0 = \mathbf{V} + jX_S\mathbf{I} = 1 + 1.8\angle 115.84^\circ = 0.2154 + j1.620 = \mathbf{1.634\angle 82.43^\circ\ pu}$$

The excitation is $\mathbf{1.634}$ **pu**.

Q4b phasor diagram: I leads V by 25.84 degrees, jXsI leans left, E0 = 1.634 at 82.43 degrees

(ii) [1] The rotor angle is $\delta = \angle\mathbf{E}_0 = \mathbf{82.42^\circ}$ (82.43° unrounded). It is close to 90° because a leading-pf generator is under-excited.

(iii) [2]

$$P_{max} = \frac{VE_0}{X_S} = \frac{1 \times 1.634}{1.8} = \mathbf{0.9078\ pu} \quad \text{at} \quad \delta = \mathbf{90^\circ}$$

Against the course: the course gives 82.42° and 0.9077 pu. Unrounded values are 82.43° and 0.9079 pu, and $1.634/1.8 = 0.90778$. The differences come only from intermediate rounding and truncation. Any of these values earns the marks.

CHECK: $P = (1 \times 1.634/1.8)\sin 82.43^\circ = 0.900$ pu, which matches 0.9×1 pu.

4c) Short-Answer Questions [9 marks]

- **Real power** comes from the part of the current that is in phase with the voltage. It is the average value of the instantaneous power. It represents energy flowing one way, from the source to the load, and it is the power that does useful work, such as heat or mechanical output. Resistors absorb it.
- **Reactive power** comes from the part of the current that is a quarter-cycle (90 degrees) out of phase with the voltage. Its average is zero: energy surges back and forth between the source and the inductors or capacitors each cycle. It does no useful work, but it still adds to the current, and so to the losses and voltage drops. Inductors absorb it and capacitors generate it.
- Both real and reactive power are scalar quantities.
- **Apparent power** is the magnitude of the complex power, whose real part is the real power and whose imaginary part is the reactive power. It is the rms voltage times the total rms current. It sets the current that generators, transformers and cables must carry, so equipment is rated in volt-amperes.

- **Copper (winding) loss:** resistive heating of the primary and secondary windings by the load current. In the simplified equivalent circuit it is modelled by the series winding resistances (the primary resistance plus the referred secondary resistance). It is proportional to the square of the current, so at constant voltage it rises with the **square of the output power**. It is small at light load and dominant at full load.

- **Iron (core) loss:** hysteresis loss (energy to repeatedly reverse the magnetisation of the steel) and eddy-current loss (currents induced in the core laminations by the alternating flux). It is modelled by the core-loss resistance in parallel across the supply. It depends on the supply voltage and frequency, through the peak flux density, and **not on the load**, so it is essentially constant with output power.
- As a result, efficiency is low at light load. It peaks at the load where the copper loss equals the iron loss.

- **Magnetising current.** Even with no load, the stator must set up the rotating magnetic field in the air gap. This needs a magnetising current, which is modelled by the magnetising reactance. An induction machine has an air gap, so its magnetising current is large, typically a quarter to over half of rated current. It is much larger than in a transformer.
- **Core-loss current.** The alternating flux causes hysteresis and eddy-current losses in the stator (and rotor) iron. These need an in-phase current component, which is modelled by the core-loss resistance.
- **Mechanical losses.** Even unloaded, the rotor must overcome bearing friction and windage (air drag, including the cooling fan). So it runs slightly below synchronous speed. The small slip induces a small rotor current that supplies this power, and the stator copper loss adds a little more.
- The no-load current is therefore mostly reactive, so the no-load power factor is very low.